

$$E := 210 \text{ GPa}$$

$$P := 1 \text{ kN}$$

$$I := 48200 \text{ cm}^4$$

$$k_y := 100 \frac{\text{kN}}{\text{m}}$$

$$L_1 := 5 \text{ m}$$

$$k_r := 1000 \text{ kN} \frac{\text{m}}{\text{rad}}$$

$$L_2 := 5 \text{ m}$$

$$u_{fixed} := \frac{P \cdot (L_1 + L_2)^3}{3 \cdot E \cdot I} = 3,2932 \text{ mm}$$

Deflection of the fixed cantilever, no hinge

$$M_h := P \cdot L_2 = 5 \text{ kN m}$$

Moment at the hinge

Case uy

$$\theta_1 := \frac{P \cdot L_1^2}{2 \cdot E \cdot I} + \frac{M_h \cdot L_1}{E \cdot I} = 0,0004$$

Rotation of the right end of segment 1

$$u_{1L} := \frac{P \cdot L_1^3}{3 \cdot E \cdot I} + \frac{M_h \cdot L_1^2}{2 \cdot E \cdot I} = 1,0291 \text{ mm}$$

Deflection of the right end of 1 from force P and moment Mh

$$u_{2P} := \frac{P \cdot L_2^3}{3 \cdot E \cdot I} = 0,4116 \text{ mm}$$

Deflection of the right end of segment 2 from force P only

$$u_{tip} := u_{1L} + \frac{P}{k_y} + \theta_1 \cdot L_2 + u_{2P} = 13,2932 \text{ mm}$$

Deflection at the tip

$$u_{1L} + \frac{P}{k_y} = 11,0291 \text{ mm}$$

Deflection at the hinge

Case Ry

$$\theta_1 := \frac{P \cdot L_1^2}{2 \cdot E \cdot I} + \frac{M_h \cdot L_1}{E \cdot I} = 0,0004$$

Rotation of the right end of segment 1

$$\theta_2 := \theta_1 + \frac{P \cdot L_1}{k_r} = 0,0054$$

Rotation of the left end of segment 2

$$u_{1L} := \frac{P \cdot L_1^3}{3 \cdot E \cdot I} + \frac{M_h \cdot L_1^2}{2 \cdot E \cdot I} = 1,0291 \text{ mm}$$

Deflection of the right end of 1

$$u_{2P} := \frac{P \cdot L_2^3}{3 \cdot E \cdot I} = 0,4116 \text{ mm}$$

Deflection of the right end of segment 2 from force P only

$$u := u_{1L} + \theta_2 \cdot L_2 + u_{2P} = 28,2932 \text{ mm}$$

Final deflection at the tip